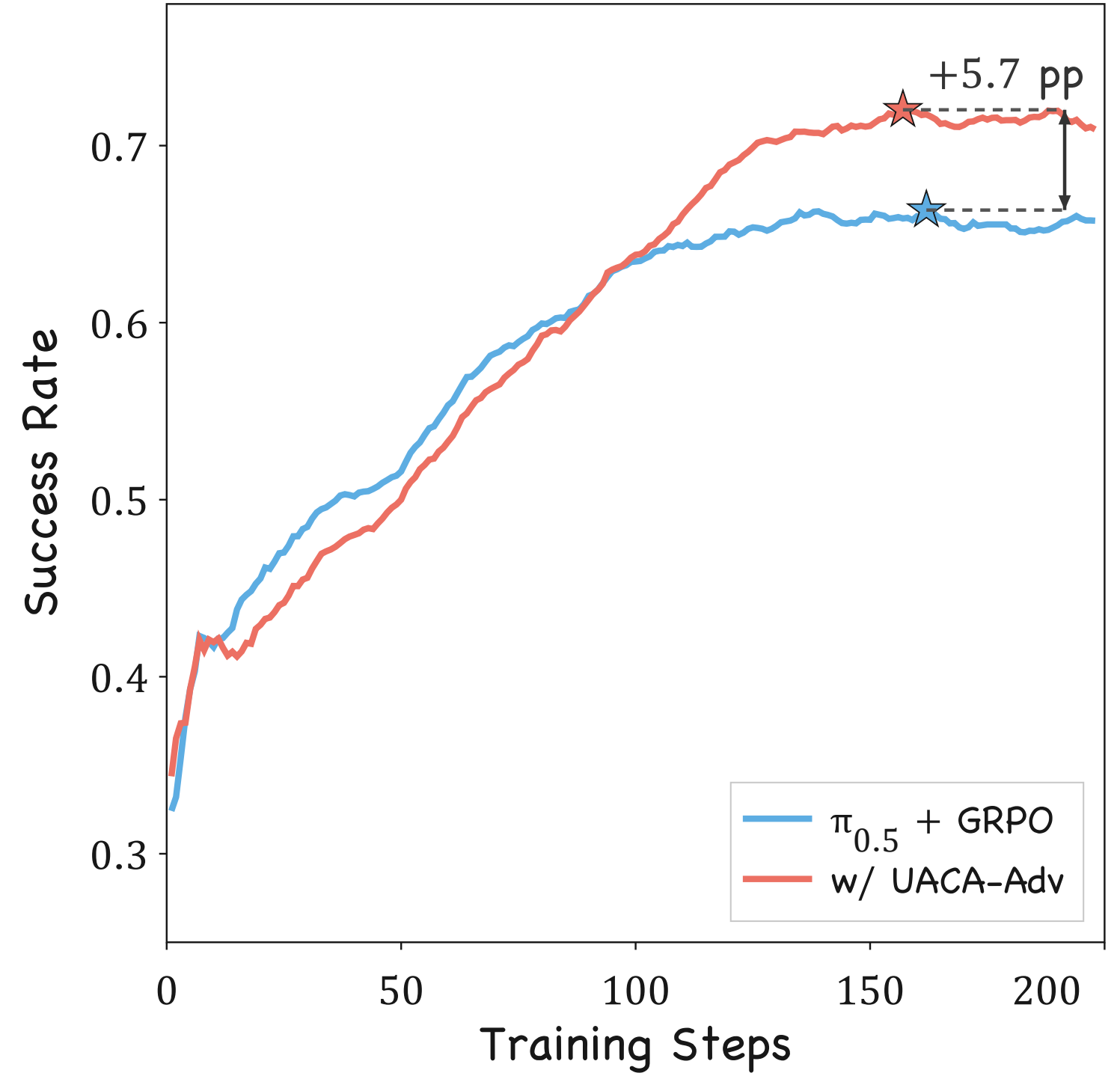
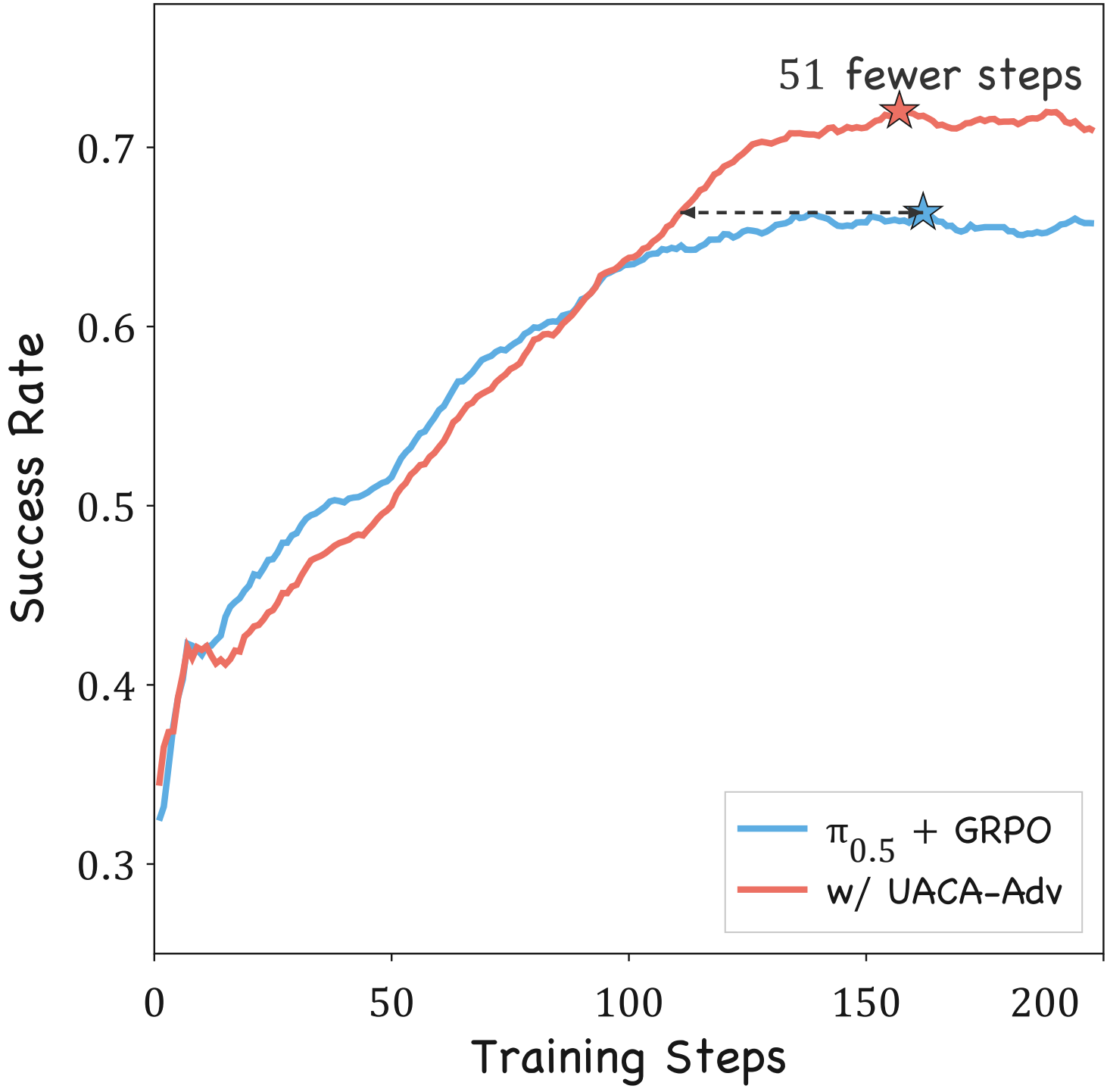
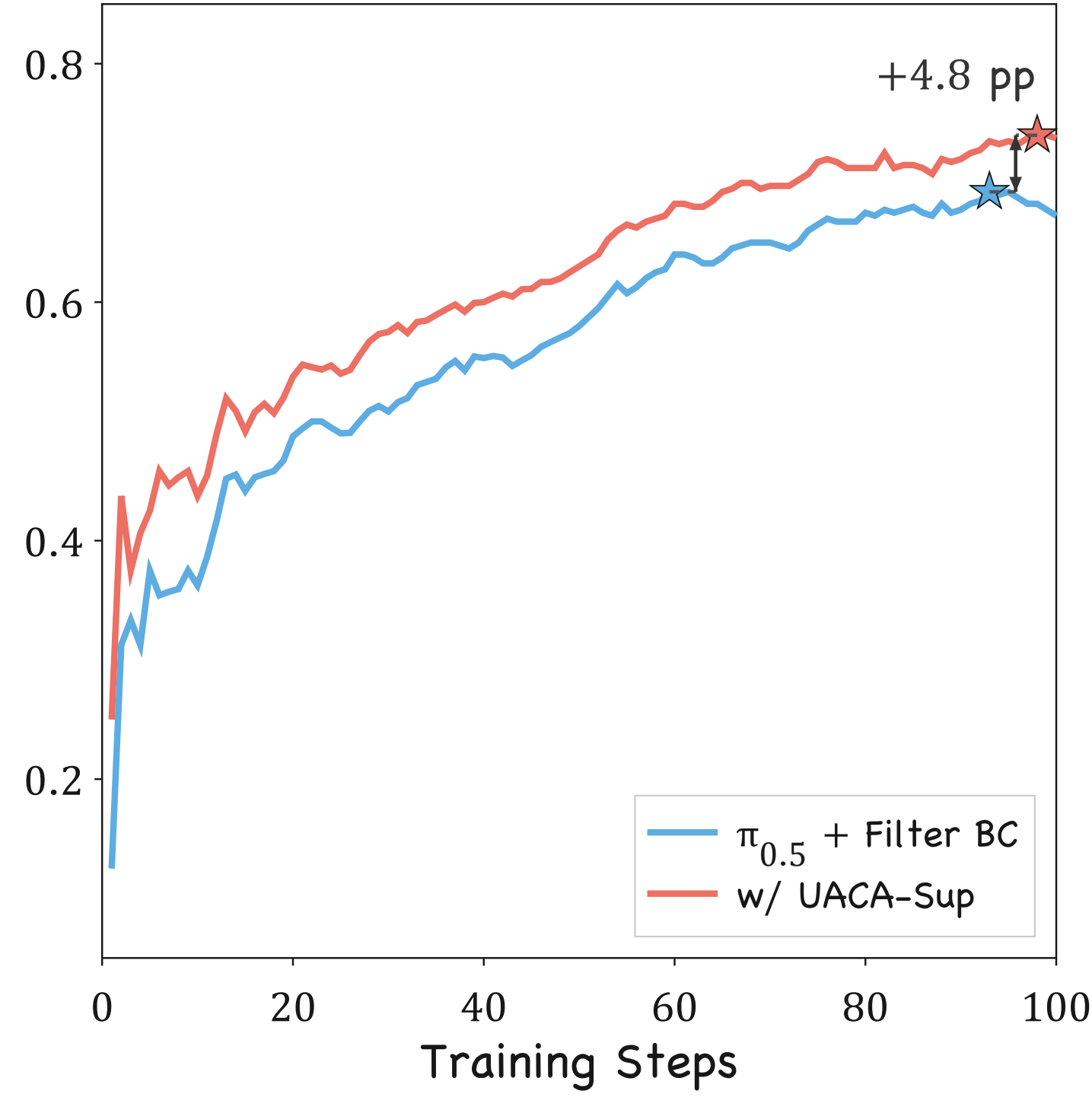
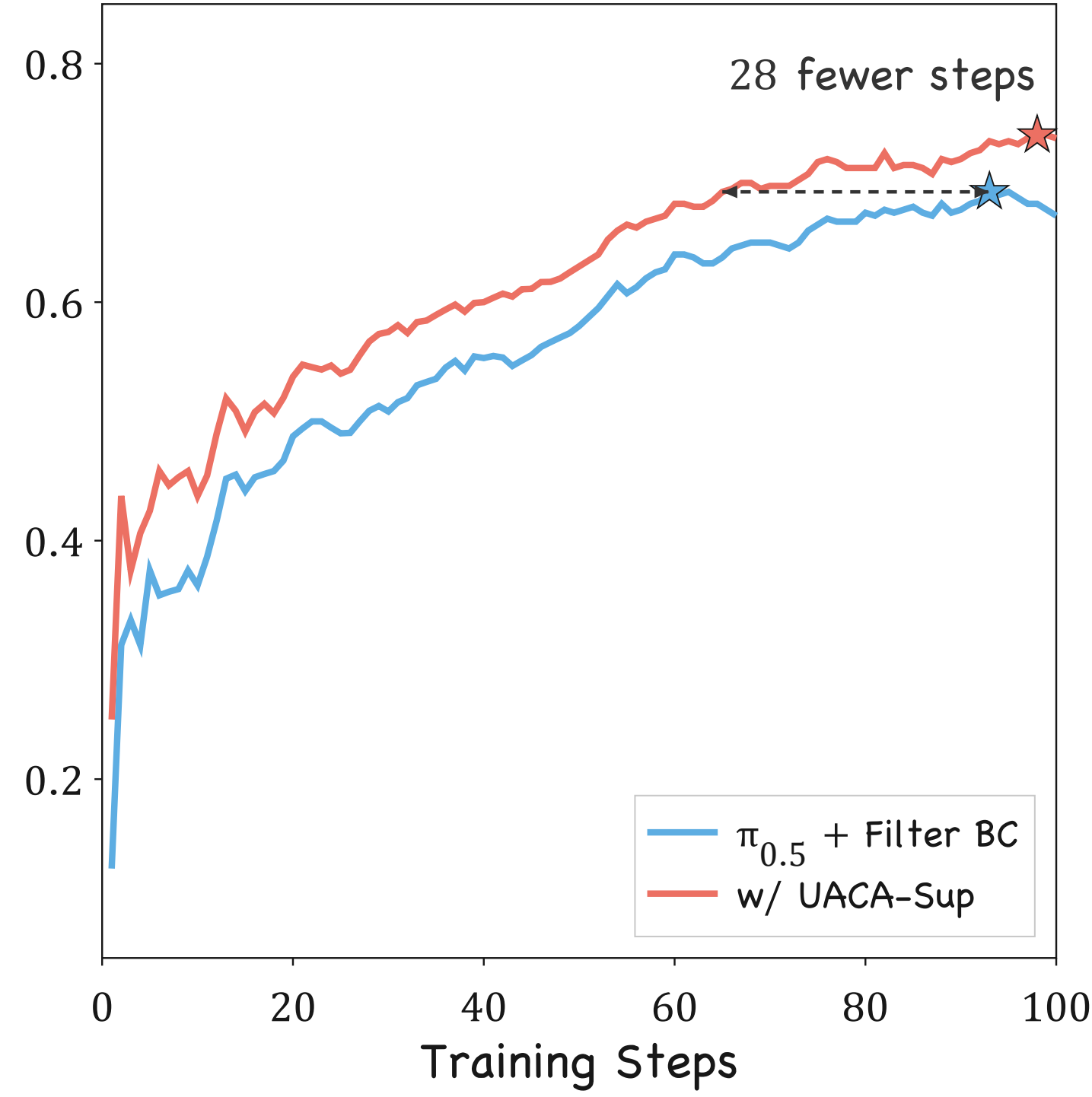
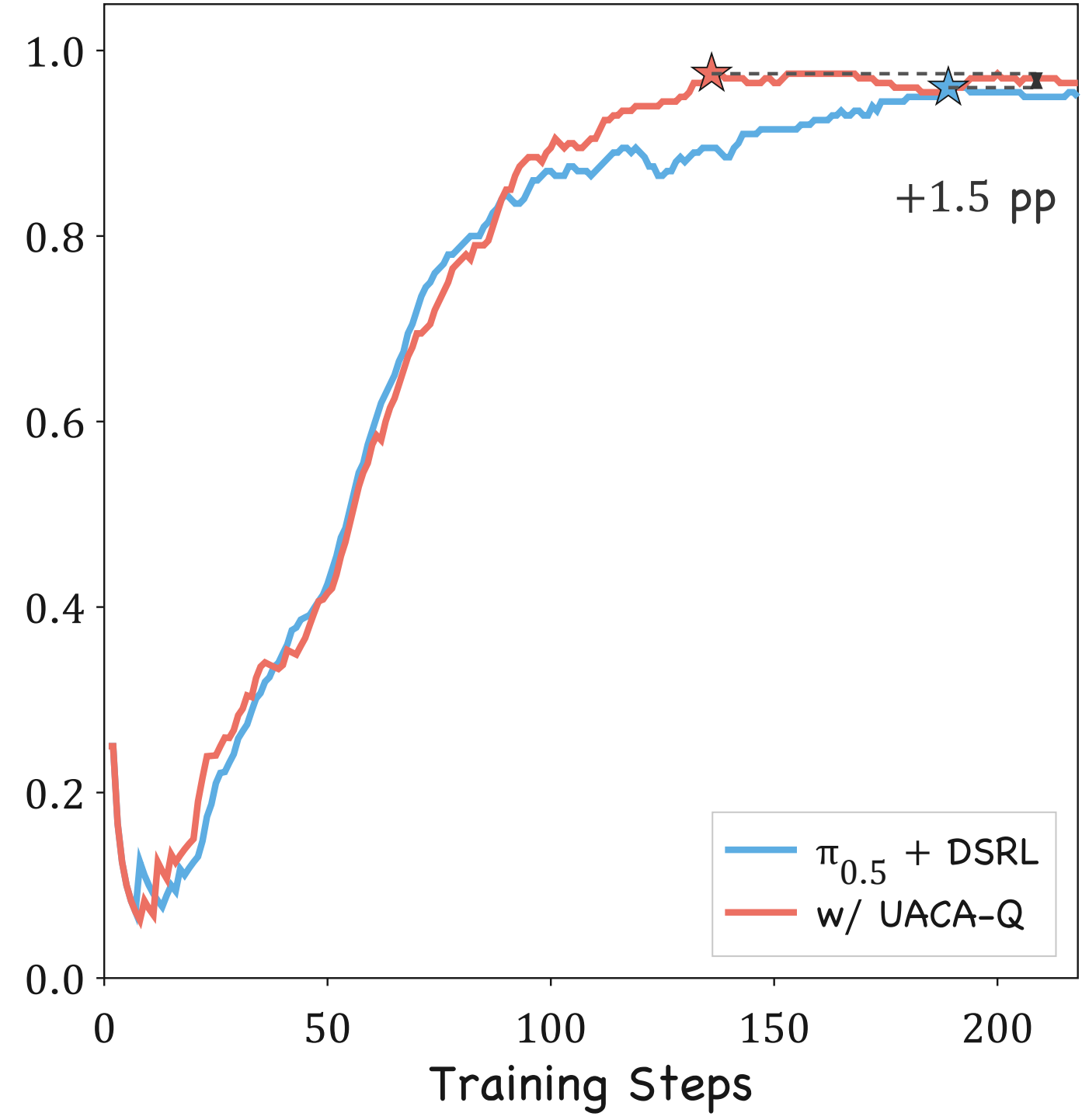
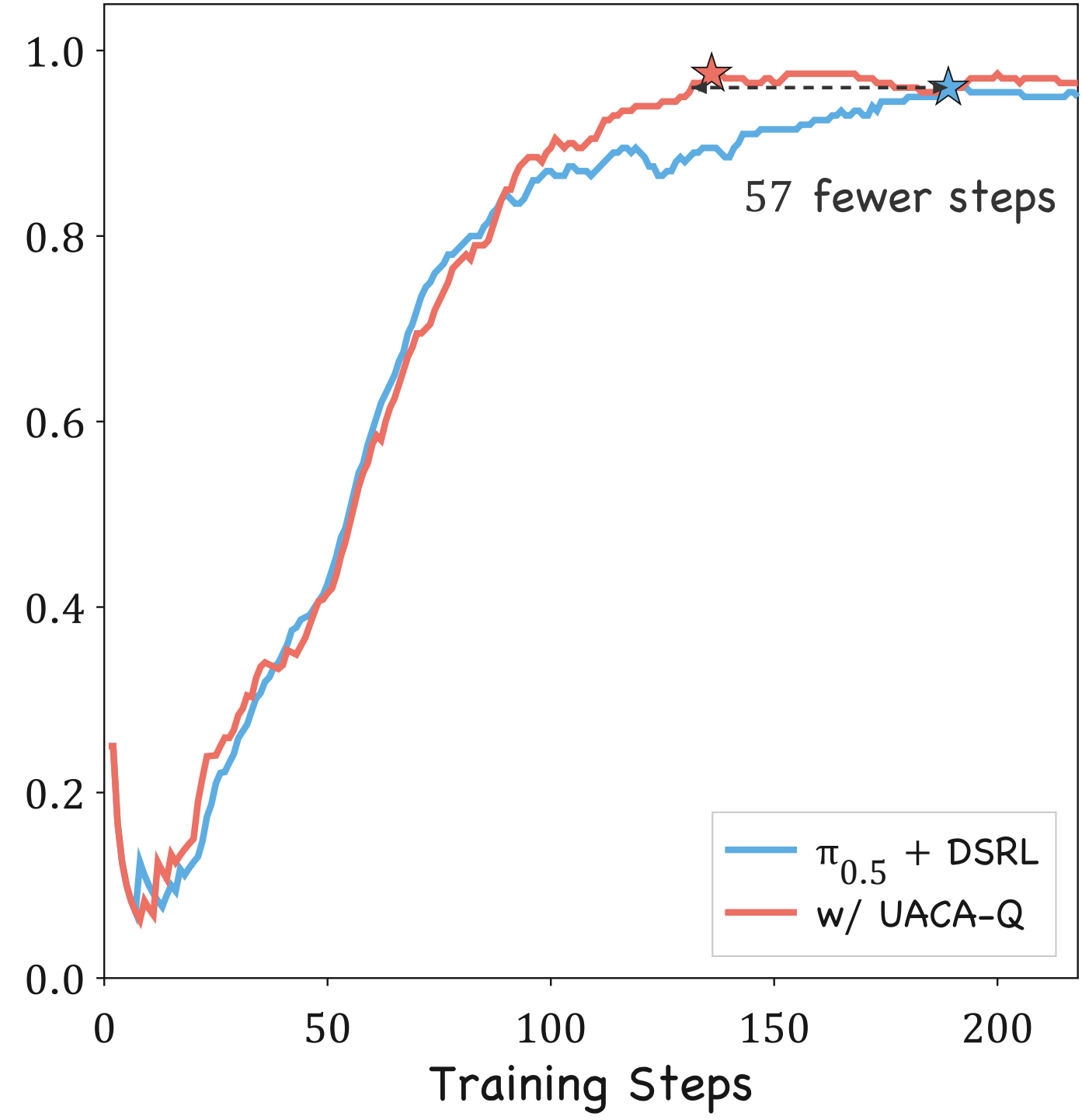

+5.7 pp
0.7
0.6
0.5
0.4
0.3
Success Rate
0
50
100
150
200
Training Steps
$\pi_{0.5}$ + GRPO
w/ UACA-Adv

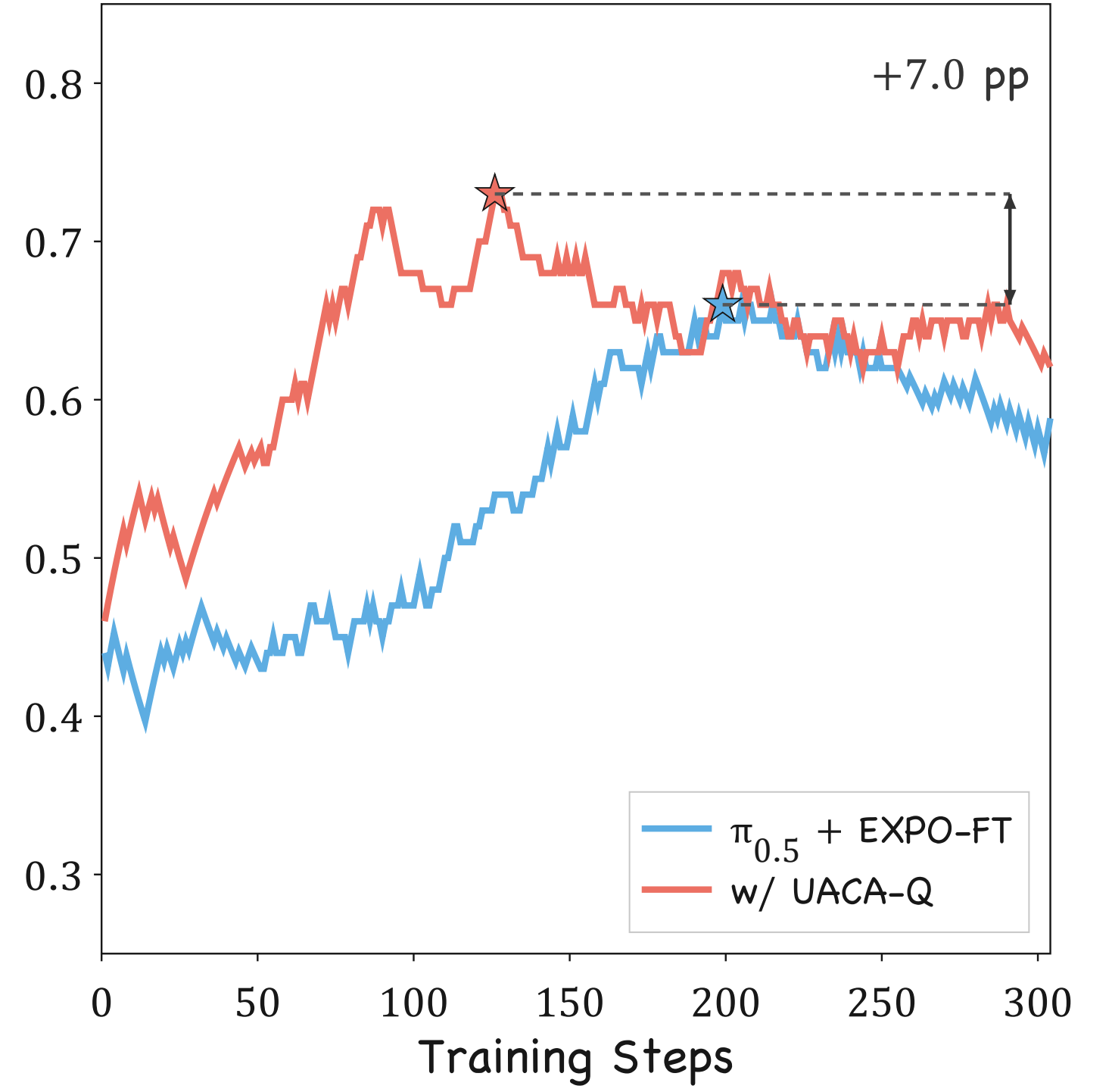

51 fewer steps
0.7
0.6
0.5
0.4
0.3
Success Rate
0
50
100
150
200
Training Steps
$\pi_{0.5}$ + GRPO
w/ UACA-Adv

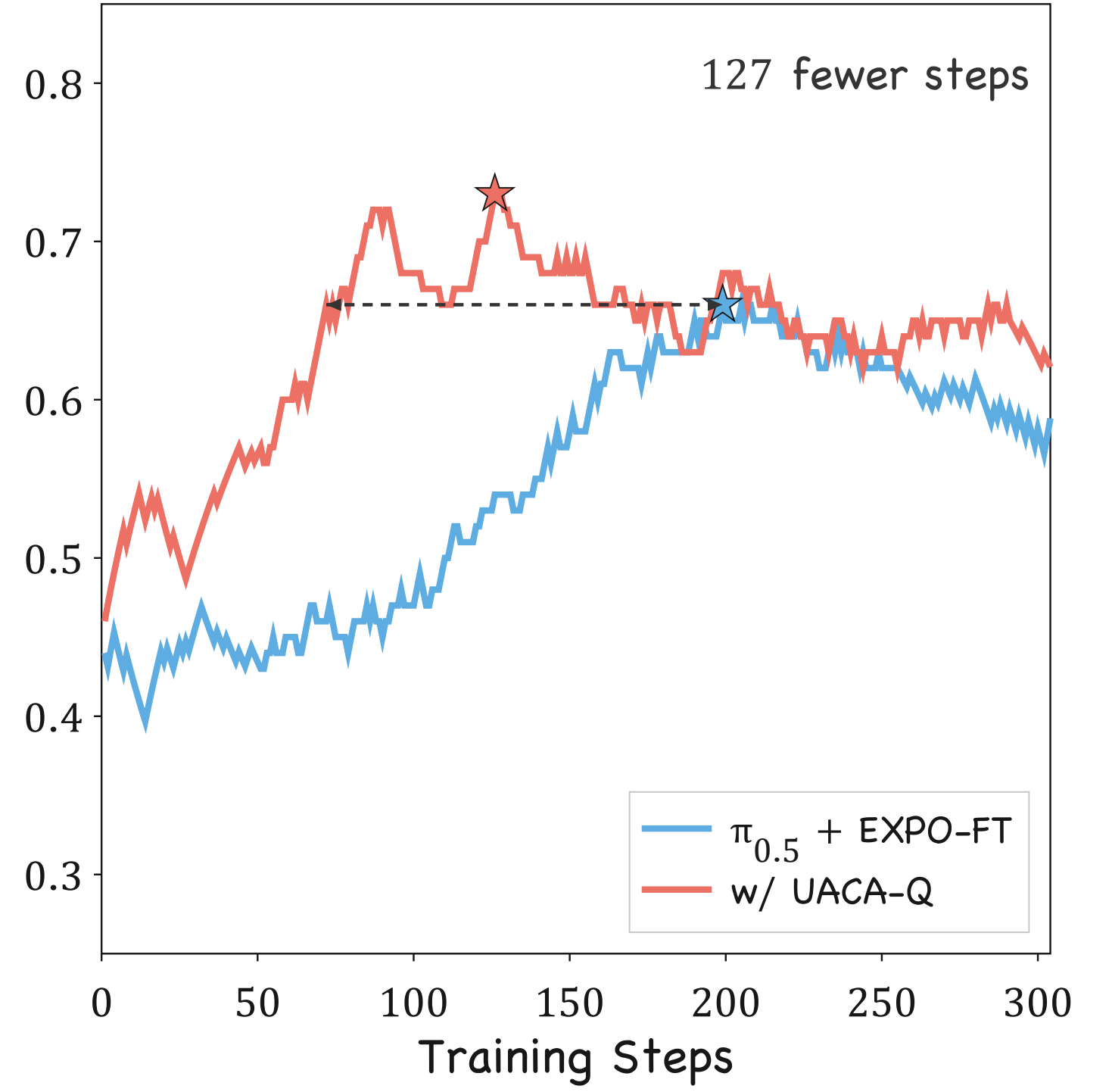

+4.8 pp
0.8
0.6
0.4
0.2
0
20
40
60
80
100
Training Steps
$\pi_{0.5}$ + Filter BC
w/ UACA-Sup

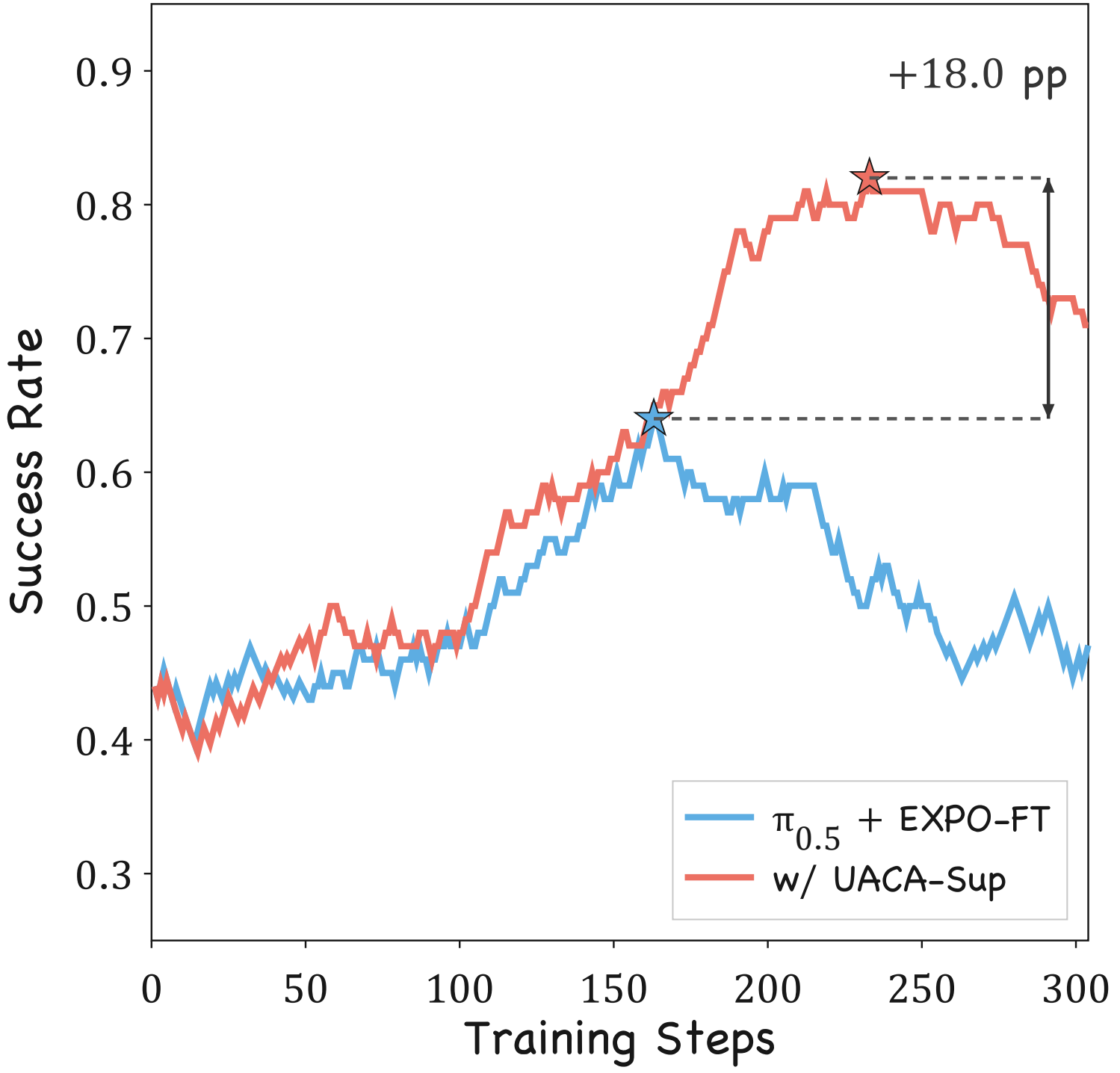

28 fewer steps
0.8
0.6
0.4
0.2
0
20
40
60
80
100
Training Steps
$\pi_{0.5}$ + Filter BC
w/ UACA-Sup

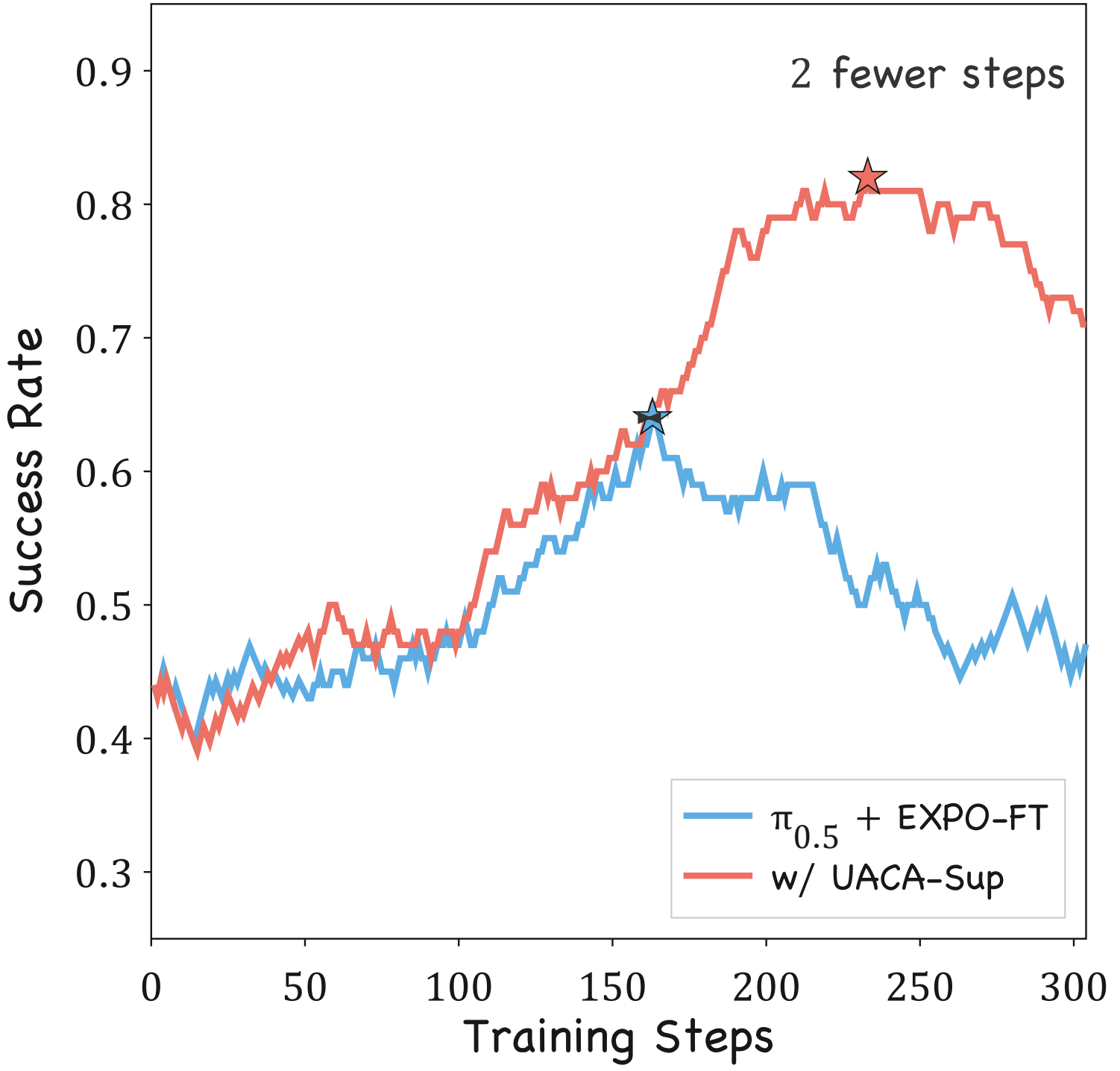

1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
Training Steps
+1.5 pp
$\pi_{0.5}$ + DSRL
w/ UACA-Q

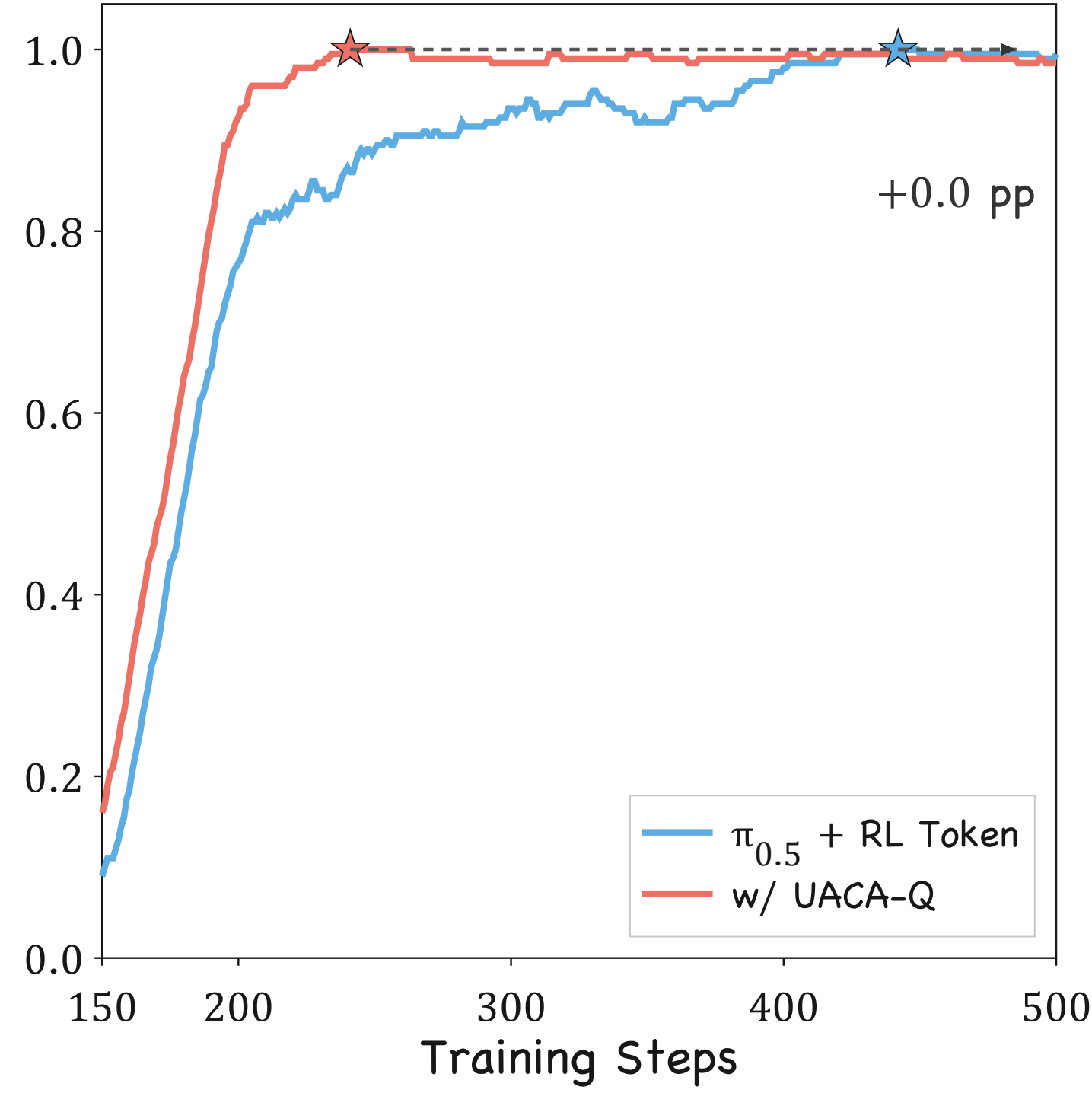

1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
Training Steps
57 fewer steps
$\pi_{0.5}$ + DSRL
w/ UACA-Q

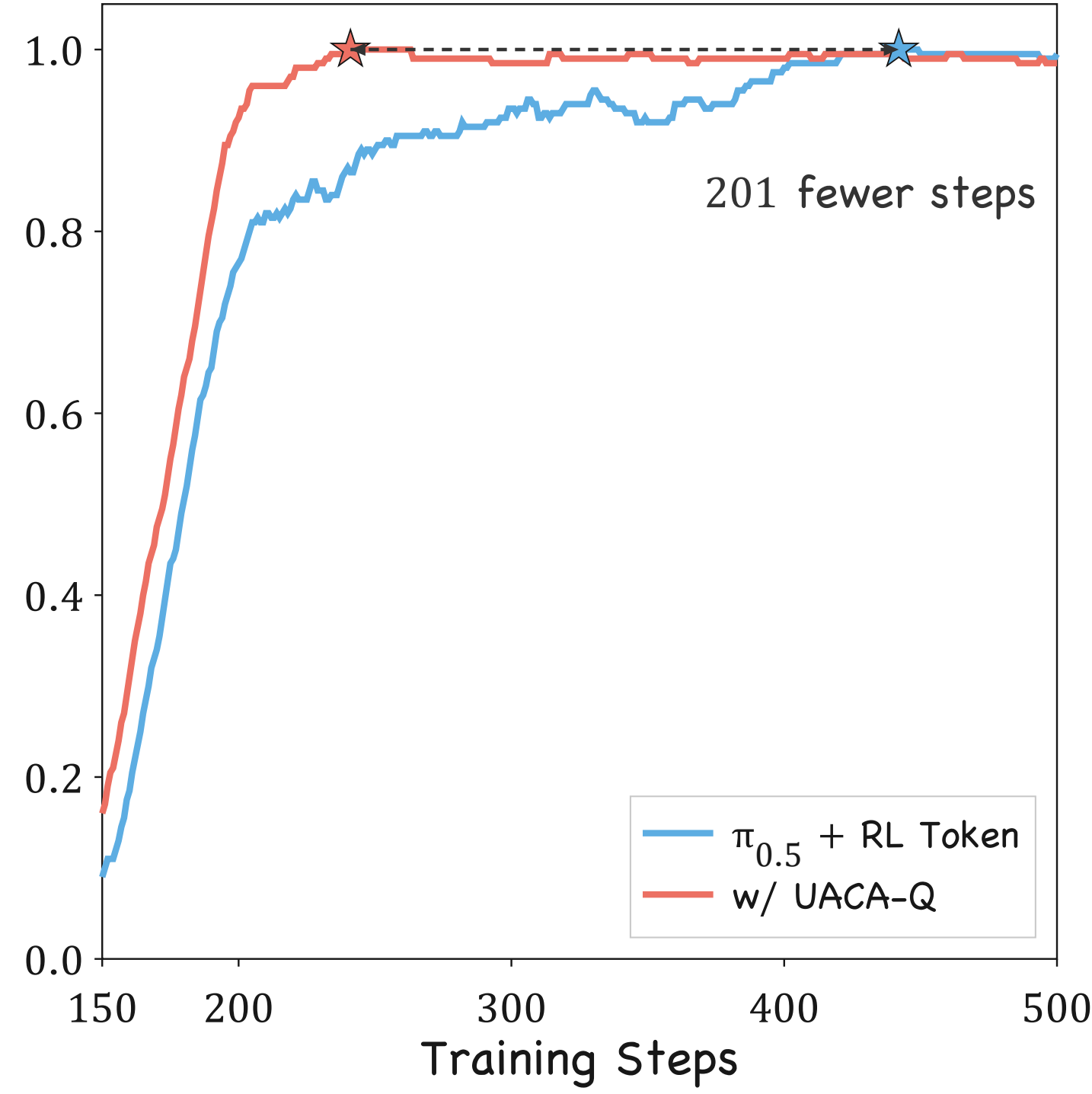

+7.0 pp
0.8
0.7
0.6
0.5
0.4
0.3
0
50
100
150
200
250
300
Training Steps
$\pi_{0.5}$ + EXPO-FT
w/ UACA-Q

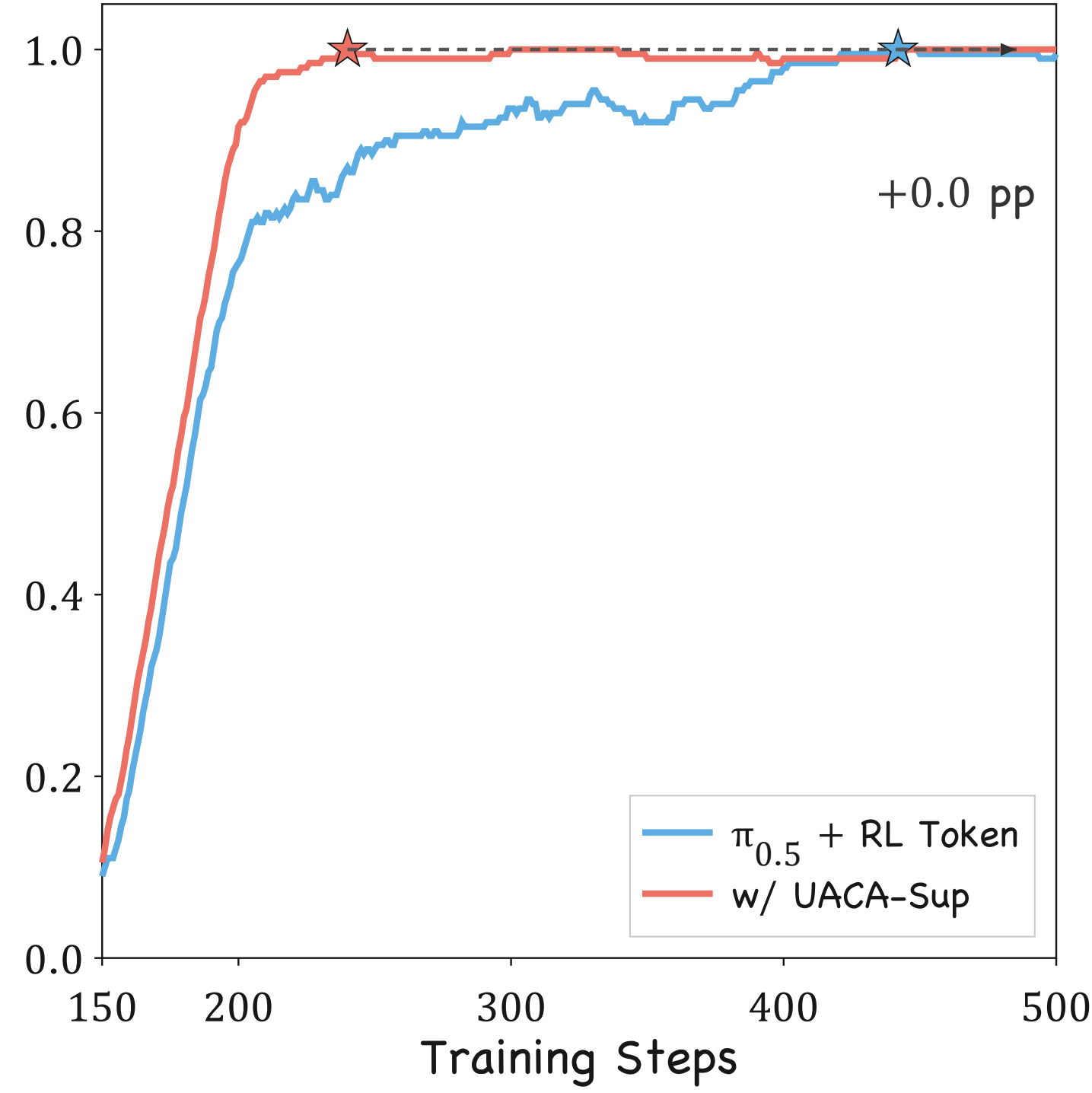

127 fewer steps
0.8
0.7
0.6
0.5
0.4
0.3
0
50
100
150
200
250
300
Training Steps
$\pi_{0.5}$ + EXPO-FT
w/ UACA-Q

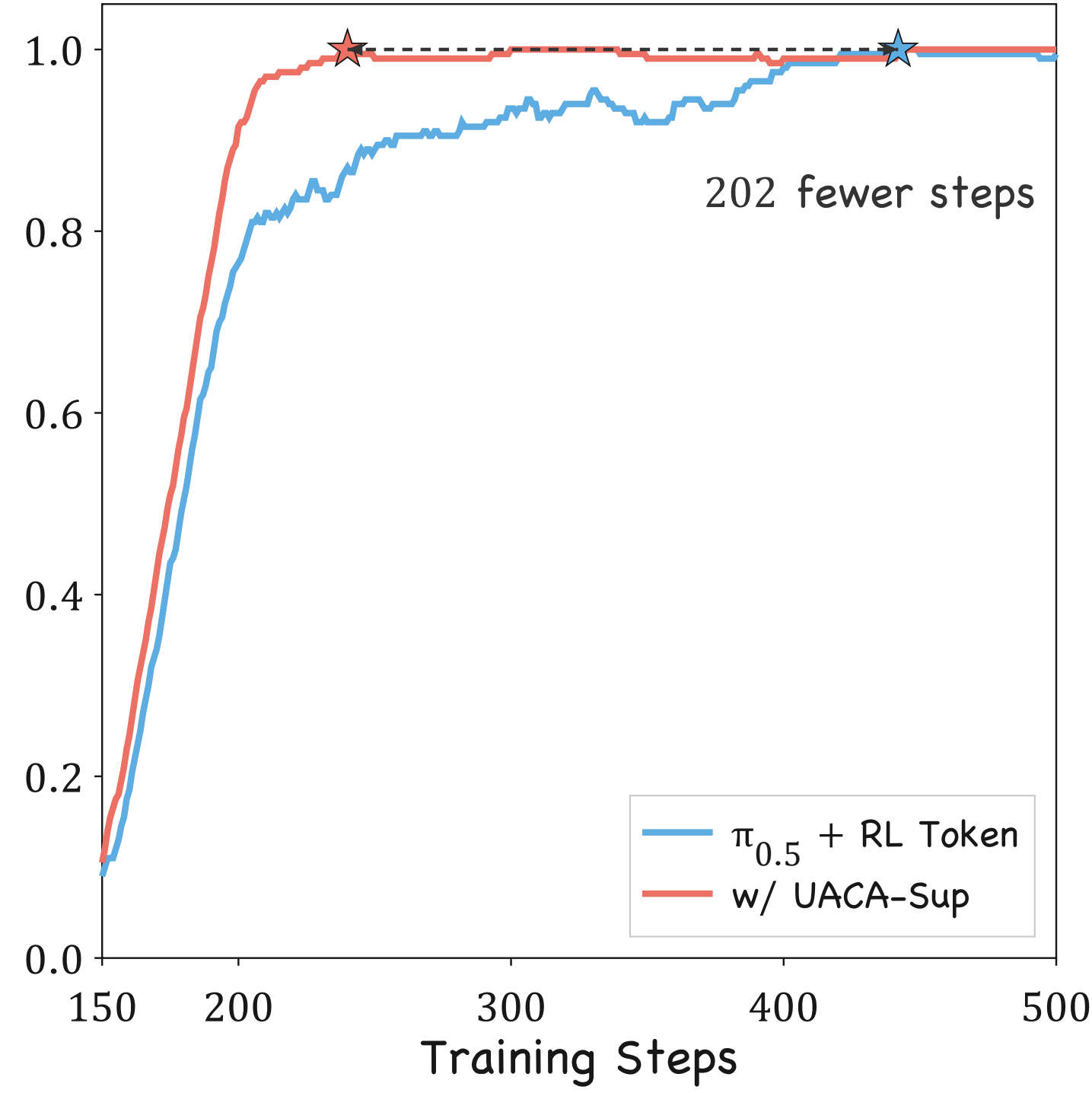

+18.0 pp
Success Rate
Training Steps
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
50
100
150
200
250
300
$\pi_{0.5}$ + EXPO-FT
w/ UACA-Sup


2 fewer steps
Success Rate
Training Steps
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
50
100
150
200
250
300
$\pi_{0.5}$ + EXPO-FT
w/ UACA-Sup


1.0
0.8
0.6
0.4
0.2
0.0
150
200
300
400
500
Training Steps
+0.0 pp
$\pi_{0.5}$ + RL Token
w/ UACA-Q


1.0
0.8
0.6
0.4
0.2
0.0
150
200
300
400
500
Training Steps
201 fewer steps
$\pi_{0.5}$ + RL Token
w/ UACA-Q


1.0
0.8
0.6
0.4
0.2
0.0
150
200
300
400
500
Training Steps
+0.0 pp
$\pi_{0.5}$ + RL Token
w/ UACA-Sup


1.0
0.8
0.6
0.4
0.2
0.0
150
200
300
400
500
Training Steps
202 fewer steps
$\pi_{0.5}$ + RL Token
w/ UACA-Sup